

Use one per cent

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find 17% of 650. Copy or complete the first step.

2. A £240 coat is reduced by 15%. What is the sale price?

3. 84 is 35% of a number. What is the number?

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find 17% of 650. Copy or complete the first step.

Answer 110.5.

2. A £240 coat is reduced by 15%. What is the sale price?

Answer £204.

3. 84 is 35% of a number. What is the number?

Answer 240.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

Answer Calculate the percentage of the original amount first, then apply that change in the context.

Use one per cent

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find 17% of 650.

6. Check this result using an inverse, estimate or known fact: 84 is 35% of a number. What is the number?

2. A £240 coat is reduced by 15%. What is the sale price?

7. Prove or explain your reasoning: Find 25% of 360.

3. 84 is 35% of a number. What is the number?

8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.

4. Solve this independently and show every stage: Find 17% of 650.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A £240 coat is reduced by 15%. What is the sale price?

10. Write and solve a similar question to: 84 is 35% of a number. What is the number?

Teacher answers and checking prompts

1. Find 17% of 650.
Answer 110.5.
2. A £240 coat is reduced by 15%. What is the sale price?
Answer £204.
3. 84 is 35% of a number. What is the number?
Answer 240.
4. Solve this independently and show every stage: Find 17% of 650.
Answer 110.5.
5. Use a second method or representation for: A £240 coat is reduced by 15%. What is the sale price?
Answer Expected result: £204. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: 84 is 35% of a number. What is the number?
Answer Expected result: 240. A valid check must be shown.
7. Prove or explain your reasoning: Find 25% of 360.
Answer 90. The explanation must show why the method works.
8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.
Answer Calculate the percentage of the original amount first, then apply that change in the context.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: 84 is 35% of a number. What is the number?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use one per cent

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Find 25% of 360.

6. Create a new example that tests the same idea as: 84 is 35% of a number. What is the number?

2. Prove the answer to this question, rather than only calculating it: Find 17% of 650.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A £240 coat is reduced by 15%. What is the sale price?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 240.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find 25% of 360.
Answer 90. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find 17% of 650.
Answer Expected result: 110.5. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A £240 coat is reduced by 15%. What is the sale price?
Answer Expected result: £204. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 240.
Answer One valid reconstruction is: 84 is 35% of a number. What is the number? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.
Answer Calculate the percentage of the original amount first, then apply that change in the context.
6. Create a new example that tests the same idea as: 84 is 35% of a number. What is the number?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.